PlantCLEF 2026 — Techniques That Worked

Cumulative progression: **F1 0.28093 (baseline) → 0.50623 (current best, Exp-16C lock 2026-05-05).**

Below are the techniques that improved leaderboard score, in roughly the order they enter the pipeline. Negative experiments are omitted.

All of the ideas listed below are proposed and performed by Alper Erten. Strategies tested by the other two teammates are excluded from this write-up; they weren't included in the final competition results submission either.

Project GitHub Repository: https://github.com/dsgt-arc/plantclef-2026

Original Proposal (Exp-1, abandoned)

- **Cross-Region Transformer with noisy-student distillation on the 212K LUCAS unlabeled set (Exp-1, original proposal)** — The original research plan was an end-to-end *cross-region transformer*: each quadrat would be sliced into region windows, each region encoded independently by a frozen pretrained ViT to produce region embeddings, and a lightweight self-attention transformer would learn inter-tile spatial / co-occurrence structure across regions and pool to a single quadrat-level multi-label prediction. A teacher-student / noisy-student loop would supervise the student on the ~212K unlabeled "pseudo-quadrat" LUCAS images that PlantCLEF provides for optional adaptation, with the teacher being a frozen ViT + simple tile-aggregation head. The expected gain was +0.01 to +0.03 macro-F1 over standard tiling-and-aggregation baselines.
    - **Why it was abandoned.** The mechanism was sound, but the supervision signal was not: the 212K LUCAS unlabeled adaptation set comes from a substantially different sampling distribution and image-statistics regime than the actual 2105-image PlantCLEF test set (different sites, different camera setups, different ecological mix). Pseudo-labels distilled on LUCAS taught the student to model a distribution the leaderboard does not score against, so any in-domain transfer was washed out by the train vs. test domain gap. Two related symptoms made the failure visible: KL/soft-distillation training over all 7806 classes caused mode collapse (the ~7742 near-zero targets per image overwhelmed the ~64 positive signals, F1 → 0); switching to BCE with negative sampling restored a working signal but only to F1≈0.143, well below what direct tiling on the test set was already achieving. The cross-region transformer idea was shelved and the project pivoted to the tiling-and-aggregation lineage that became Exp-2 onward.

Test Image Preprocessing

- **JPEG round-trip preprocessing (Exp-2)** — After Lanczos-resizing each test image, re-encode it through PIL JPEG at quality=85, 4:2:2 chroma subsampling, then decode back to pixels before tiling. The DINO backbones were pretrained on JPEG-compressed natural images, so passing the test images through one round of compression matches the pixel-statistic distribution they expect. Cheap, no parameters to tune, downside is zero. Δ ≈ +0.002 standalone, compounds with downstream stages.

- **Multiscale non-overlapping tiling, scales 4+5+6 = 77 tiles (Exp-7)** — A single 4×4 grid on a 3000×3000 quadrat fragments individual plants and mixes species. Instead, tile each image at three grid sizes (4×4, 5×5, 6×6), giving 77 non-overlapping 518×518 tiles at three effective source resolutions (~747, 600, 500 px). The downstream `max_prob` aggregator can then pick whichever scale best isolates each species. Coarser scales (1×1, 2×2, 3×3) hurt because whole-image crops produce confident wrong labels; 7×7 is diminishing return. Δ ≈ +0.028 over 4×4 alone.

Aggregation & Fusion

- `max_prob` **per-tile aggregation, T = 1.5 (Exp-4)** — For each species class, take the maximum softmax probability across the 77 tiles as the image-level prediction (with temperature T=1.5 sharpening per-tile peaks). Noisy-OR multiplies tile noise; mean/quantile pooling dilutes strong signals. Max preserves the single-best tile evidence per class independently, and is the only aggregator whose F1 is invariant to tile count — the precondition that lets multiscale work. Δ ≈ +0.039.

- **Temporal** `prior_blend` **across location visits (Exp-4)** — Each physical plot in the test set is photographed multiple times. Build a location prior `P_loc(c) = max over visits of P_visit(c)`, then blend with the current image's prediction: `P_final = (1−α)·P_loc + α·P_image`. Applied only to multi-visit quadrats (~76% of the test set). Lets a weak current photo borrow species evidence from earlier visits at the same plot. Δ ≈ +0.022.

- **Per-model α (Exp-11)** — Each branch of the ensemble gets its own α: ViT+kNN α=0.15 (already has external evidence from kNN, so leans on the location prior), ConvNeXt α=0.30 (no kNN arm, trusts its own per-image prediction more, which preserves architecture diversity for the ensemble). Δ ≈ +0.001.

- **Per-source** `sim_w` **aggregator on RNNB + α=0.30 (Exp-14, Stage 4)** — RNNB sites (saltmarsh / coastal dunes in SW France) show year-to-year species turnover: a lush May visit looks nothing like the bare-sand January visit at the same plot, but `max`-pooled prior was making them produce identical predictions. Replace with a per-image prior aggregator weighted by cosine similarity between the current image embedding and each historical sister visit: similar visits contribute fully, dissimilar ones collapse out. Restricted to RNNB only — applying it globally cratered alpine species counts (10.0 → 4.85). Alpine sources keep `max`. Δ +0.00982.

Classifier Model Ensembling (ViT-L-DINOv2 + ConvNext-L-DINOv3)

- **ConvNeXt-Large DINOv3 finetune, 2-phase (Exp-10)** — `convnext_large.dinov3_lvd1689m` (196M params). Phase 1: freeze the backbone, train only the 7806-way classifier head with Adam, lr=0.01 for 100 epochs (cheap, stable head warmup). Phase 2: unfreeze everything and finetune end-to-end from the Phase 1 EMA weights with AdamW, lr=3e-5, layer-wise lr decay 0.8, drop_path=0.3 (higher reg for the larger model), RandAugment + mixup + cutmix, EMA decay 0.9998, 100 epochs on 4×H200. Provides the second ensemble member needed for architecture diversity.

- **ViT-L Phase 2v3 cooldown (Exp-13)** — `vit_large_patch14_reg4_dinov2.lvd142m` finetuned in two stages following the same head-only → full-finetune protocol, then a 10-epoch cooldown at lr=2e-5 on combined train+val+test data on top of Phase 2. Replaces the earlier ViT-B as the production ViT backbone. Standalone +0.011 F1.

- **ConvNeXt + ViT probability-level ensemble, w=0.77 (Exp-11, "Approach A")** — Run the ViT branch with kNN blending first to get the ViT-side per-image probabilities, then linearly mix in ConvNeXt: `P =`

`0.77·P_vit+knn + 0.23·P_cnxt`. Order matters: blending logits *before* kNN breaks the embedding↔logit alignment the FAISS index assumes (it was built on ViT embeddings only) and loses ~0.02. Both branches use the same 4+5+6 scale scheme — letting ConvNeXt use its own preferred 3+4+5+6+7 introduced disagreements ViT couldn't compensate for. Δ ≈ +0.020.

Retrieval Model Ensembling (FAISS)

- **ViT-only FAISS kNN blend (Exp-9)** — Build an exact-search FAISS `IndexFlatIP` over ~860K geo-filtered ViT training-set CLS embeddings (768-d, L2-normalised so inner product = cosine). For each test tile embedding, retrieve the k=20 nearest training neighbours, softmax the similarities with τ=0.07 to get neighbour weights, and aggregate by class label to form a retrieval distribution `p_knn`. Blend per-tile *before* `max_prob` aggregation: `p_tile = β·softmax(logits/T) + (1−β)·p_knn`. The classifier softmax is biased toward common species (long tail under-trained); the retrieval arm is frequency-agnostic and supplies complementary "look-up" evidence for rare species clusters. Δ ≈ +0.017.

- **Differentiated β: β_multi=0.80, β_single=1.00 (Exp-9)** — Inject kNN only on multi-visit quadrats. Single-visit images don't have the temporal prior to absorb retrieval noise; using kNN there cost up to −0.08 F1. Multi-visit quadrats can tolerate β=0.80 (20% kNN weight) because the location prior smooths out noisy retrievals.

Ecological Masking

- **Aquatic-species habitat-impossibility mask (Exp-14, Stage 12)** — Hand-curate 35 fully-aquatic genera (Posidonia, Zostera, Cymodocea, Lemna, Potamogeton, etc.) and 6 aquatic specific epithets (oceanica, submersa, fluitans, natans, …). Mark any of the 7806 species whose Latin binomial matches as habitat-impossible (91 species), and zero those probabilities post-fusion across all sources. Triggered by inspection: Posidonia oceanica (a marine seagrass) was the most-predicted species on dry-soil RNNB sand-dune quadrats. Pure precision improvement, zero parameters, ecologically defensible. Δ +0.00162.

- **Per-source biogeographic mask on alpine sources (Exp-14, Stage 13)** — On CBN-PdlC and CBN-Pla (alpine summit plots only), zero 5 hand-vetted species that survived Stage 12 (no aquatic match) but cannot occur at alpine elevation: tropical C4 carpetgrass (Axonopus compressus), coastal sea-thrift (Armeria maritima), Mediterranean coastal annuals, etc. Other sources are unchanged. Δ +0.00039 LB but proxy-positive on the full cohort; adopted as new best.

Source-Aware Admission & Demotion

- **N-floor admission with confidence-tiered ratio gate, "N3_r0.5" (Exp-16)** — On count-bound sources where the model under-predicts (GUARDEN, LISAH, OPTMix, RNNB), if fewer than 3 species pass the standard threshold=0.100, admit additional candidates up to 3 — but only those whose probability ratio to the already-admitted top-1 is ≥ 0.5 (so we admit confident near-ties, not random low-confidence noise). Doesn't touch sources that already predict enough species (alpine). Largest single-mechanism gain since Stage 4. Δ +0.00557.

- **Habitat-fit demotion using external training-data geography (Exp-16)** — For each (test image, candidate species) pair, compute a habitat-fit factor in [β, 1]. Two components multiplied together: (1) **geo score** — count training observations of that species within 200 km of the source's refined regional centroid; if ≥ n_near_min (40) the species is unpenalised, if 0 it's demoted to β, linear in between; (2)

altitude score — penalise species whose training-data median altitude lies outside the source's expected band. Multiply into the species probability *before* the N-floor admission. Uses external information (PlantCLEF2024 training-data lat/lon/alt) the model never saw at inference. Cosmopolitans (high n_near everywhere) untouched; range-restricted species observed only in distant regions get demoted to β. Δ +0.02361 standalone, final +0.03674 combined with admission retune.

- **Per-source β override on alpine sources (Exp-16C, NEW MECHANISM)** — The single global β was demoting legitimate alpine species too hard, because alpine quadrats were pinned at the topk-10 cap and any extra demotion just dropped real positives below threshold. Override β to 0.45 on CBN-PdlC and CBN-Pla (vs global β=0.15). Tested OPTMix per-source β too — no-op because OPTMix's "out-of-place" species (Plantago subulata-style) have high local n_near and aren't hit by the multiplicative knob. Δ +0.00065 LB, locks final F1=0.50623.

- **Habitat-fit retune at new ViT-L+ConvNeXt-P backbone (Exp-16C)** — When the production backbone changed, the habitat-fit hyperparameters needed re-tuning: n_near_min 30→40, β_global 0.10→0.15. R=200 km, k=10, thr=0.085 unchanged. Final lock 2026-05-05.

Final Submission

- **Fixed global threshold = 0.085 + topK = 10 + SW Europe geo mask** — Hard-threshold per-class probabilities, then cap at 10 species per image. Adaptive per-image thresholds (Exp-6: max-gap, relative-ratio, elbow) all lost to a well-tuned fixed threshold. The geo mask zeroes ~3000 of the 7806 classes that don't occur in southwest Europe. Threshold value drifted across experiments (0.04 → 0.10 → 0.085) only because the probability scale shifts when the pipeline re-blends — the *rule* (fixed global threshold) is unchanged.